MIL-C-25/3E
21 June 1968
SUPERSEDING
MIL-C-25/3D
29 September 1960

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PAPER-DIELECTRIC, DIRECT CURRENT (HERMETICALLY SEALED IN METAL CASES), STYLES CP40 AND CP41

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-25.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

REQUIREMENTS:

Dimensions: See figure 1 and table I.

Design and construction: Capacitors shall be of the design, construction and physical dimensions as specified in figure 1 and table I.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC voltage rating: See table I.

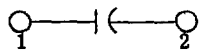
Operating temperature range: -55° to +85° C.

Vibration grade: 1 (10 to 55 Hz, incl.)

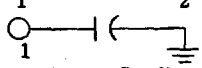
Terminal:
Symbol B - - - - Solder lug (nonremovable).
Symbol C - - - - Threaded stud and nuts.

Circuit diagram:

Symbol 1 (CP41) - - - -



Symbol 2 (CP40) - - - -



Voltage derating with temperature: See figure 2.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
Terminal to terminal - As specified in table III.
Terminal to case (when case is not a terminal) - As specified in table III. Capacitors shall withstand specified voltages with no permanent damage or open- or short-circuiting.

Insulation resistance: Method 302 of MIL-STD-202.
At 25° C:
Terminal to terminal - As specified in table IV.
Terminal to case (when case is not a terminal) - Shall exceed 3,000 megohms.
At 85° C:
Terminal to terminal - As specified in table IV.

Capacitance: Method 305 of MIL-STD-202.
Measured capacitance shall be within the tolerance specified in table I.

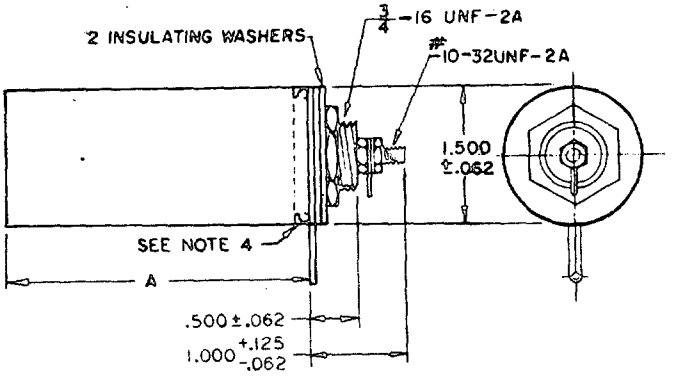
Dissipation factor: When measured as specified in MIL-C-25, shall not exceed 1 percent.

Barometric pressure (flashover): Method 105 of MIL-STD-202.
Test condition - B (50,000 feet).
Test potential - 125 percent of rated dc voltage applied for a minimum of 1 minute, without visible damage, external flashover, or open- or short-circuiting.

Vibration (grade 1, 10 to 55 Hz, incl):
Method 201 of MIL-STD-202.
Test duration - 3 hours (60 minutes in each direction.
Measurements - An electrical signal of 1±0.2 kHz, at a level of 1±0.5 volts applied during the last 30 minutes in each direction. There shall be no momentary arcing or breakdown, open- or short-circuiting, or mechanical damage.

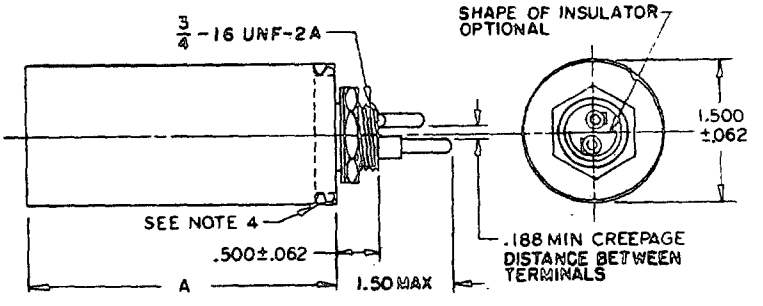
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STYLE CP40

INCHES	MM
.062	1.57
.064	1.63
.125	3.18
.188	4.78
.500	12.70
1.000	25.40
1.500	38.10



STYLE CP41

NOTES:
1. See table I for additional dimensions.
2. All dimensions in inches.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. End shape optional, shown by dotted lines.
5. Shape of terminal B (drilled, punched, forked, or slotted (optional).
6. Style CP40 shall be supplied with 1 ground lug, 2 insulating washers (1 washer extruded), 1 lockwasher, and 1 mounting nut. Available in terminals B and C. In addition, capacitors having terminal C shall be supplied with the following items of terminal hardware: 2 nuts, 1 lockwasher, and 1 terminal lug capable of accomodating a bare wire of .064 dia(No. 14 AWG).
7. Style CP41 shall be supplied with mounting nut and lockwasher. Available in terminal B only.

FIGURE 1. Styles CP40 and CP41 capacitors.

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TABLE I. Styles CP40 and CP41 characteristics and dimensions.

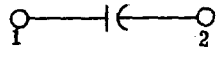
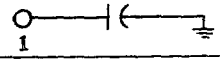
Type designation 1/	DC voltage rating (see fig. 2)	Capacitance	Terminal	Capacitance tolerance 2/	Characteristics	Vibration grade	Case dimension A±0.062	
							Characteristic E	Characteristic F
	Volts	μf						
CP4----F105K1	600	1	B, C	K	E, F	1	2.375	2.000
CP4----F205K1	600	2	B, C	K	E, F	1	3.375	2.875
CP4----F405K1	600	4	B, C	K	E, F	1	5.250	4.500
CP4----G105K1	1,000	1	B, C	K	E, F	1	3.375	2.875
CP4----G205K1	1,000	2	B, C	K	E, F	1	5.250	4.500
CP4----H254K1	1,500 3/	0.25	B, C	K	E, F	1	2.375	2.000
CP4----H504K1	1,500	0.5	B, C	K	E, F	1	2.875	2.375
CP4----H105K1	1,500	1	B, C	K	E, F	1	4.500	3.875

1/ Complete type designation shall include additional symbols to complete style designation and to indicate terminal, circuit, and characteristic.
2/ Capacitance tolerance in percent: K, ± 10.
3/ For derating purposes, watt-second group I is limited to capacitors having dc voltage ratings of 1,000 volts or less. Watt-second group II includes capacitors having a watt-second rating of 0.5 or less which are not included in watt-second group I.

TABLE II. Millimeter equivalents of decimal inches.

In	mm	In	mm
0.062	1.57	3.375	85.73
2.000	50.80	3.875	98.43
2.375	60.33	4.500	101.60
2.875	73.03	5.250	127.00

TABLE III. Dielectric withstanding voltage test.

Circuit symbol	Circuit diagram	Test	Test connections	Test voltage (percent of rated dc voltage)	Time test applied (minutes)
1		Terminal to terminal	1 to 2	200 1/	1 2/
		Terminal to case	1 to 2 to case	400 5/	1 3/ 4/
				200 + 1,000 6/ volts	1 3/ 4/
2		Terminal to terminal	1 to case	200	1 2/

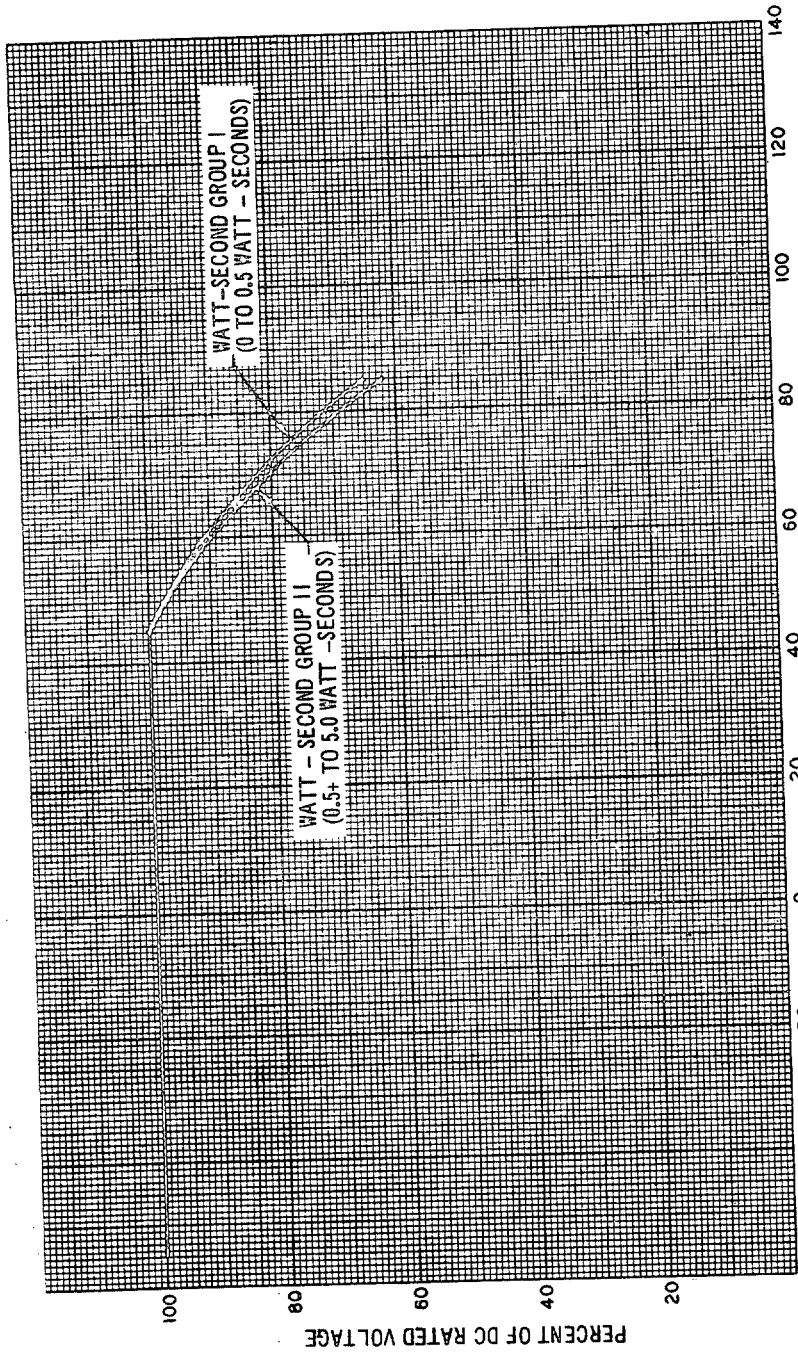
1/ 175 percent of rated dc voltage after temperature-and immersion-cycling and moisture resistance tests.
2/ For quality conformance inspection and at the option of the supplier, either 200 percent of rated dc voltage shall be applied for 15 seconds or 250 percent of rated dc voltage for not less than 1 second.
3/ For quality conformance inspection, the period of test-voltage application shall be 1 second.
4/ For quality conformance inspection, applications may be made between each terminal individually and the case, at the option of the supplier.
5/ For capacitors having dc voltage ratings of 600 volts and less.
6/ For capacitors having dc voltage ratings greater than 600 volts.

TABLE IV. Terminal to terminal insulation resistance measurements.

Capacitance values	Minimum insulation resistance 1/	
	Characteristic E	Characteristic F
	At 25° C	At 25° C
0.33 microfarads and less - - - - - -	6,000 megohms	4,500 megohms
greater than 0.33 microfarads - - - -	2,000 megohm-microfarads 2/	1,500 megohm-microfarads 2/
	At 85° C	At 85° C
greater than 0.033 microfarads - - -	20 megohm-microfarads 2/	15 megohm-microfarads 2/

1/ A potential equal to the dc rated voltage or 500 volts dc, whichever is less, shall be applied.
2/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

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NOTE: Permissible operating voltages, while based on incomplete data, are the capacitor suppliers' best estimate to provide a life expectancy of 8,800 hours of continuous operation at higher ambient temperatures. Longer life can be expected by operation at voltages lower than indicated on the curve; for example, a life expectancy of 44,000 hours may be obtained by operation at 70 percent of the voltage determined by use of this curve. Also, a life longer than 8,800 hours may be expected at the voltage determined by use of this curve if the high ambient temperature prevails for only a portion of the whole operating time.

FIGURE 2. Voltage derating for ambient temperature (characteristics E and F).

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Salt spray (corrosion):
Method 101 of MIL-STD-202, test condition B (48 hours), with no evidence of corrosion or obliteration of marking.

Temperature cycling:
Method 102 of MIL-STD-202, test condition D.

Immersion:
Method 104 of MIL-STD-202, test condition C, except that the duration of each immersion shall be 30 minutes, and change between solutions shall be made in not more than 3 seconds.
Measurements after final cycle:
Dielectric withstanding voltage, at 25°C:
Terminal to terminal - As specified in table III.
Terminal to case (when case is not a terminal) - As specified in table III.
Insulation resistance, at 25° C:
Terminal to terminal - Not less than 30 percent of the value specified in table IV.
Terminal to case (when the case is not a terminal) - Not less than 1,500 megohms.

Terminal strength:
Method 211 of MIL-STD-202, test conditions A and E.
Terminal B:
Test condition A (pull), applied force - 5 pounds.
Test condition E, torque - 2 pound-inches.
Terminal C:
Test condition A (pull), applied force - 10 pounds.
Test condition E, torque - 10 pound-inches.

Moisture resistance:
Method 106 of MIL-STD-202.
Polarization voltage - 100 volts.
Final measurements:
Dielectric withstanding voltage and insulation resistance requirements - As specified in immersion.

Seal: Method 112 of MIL-STD-202, test condition A, except that bath temperature shall be $+85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C.
Liquid-filled capacitors - 10 minutes at $+85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C on each of two opposite sides, with no evidence of leakage.

Low temperature and capacitance change with temperature:
Low temperature -
Conditions: -55° ± 3° C for 48 ± 4 hours with rated dc voltage applied.
Capacitance change with temperature:
Characteristic E - Not to exceed ± 10 percent.
Characteristic F - Not to exceed +10, -30 percent.

Life: Method 108 of MIL-STD-202, test condition B ($250\ ^{+8}_{-0}$ hours).
Electrical load:
Group IA (0.08 watt-second and less), excepting 1,000 volt capacitors - 140 percent of dc rated voltage.
Group IB (greater than 0.08 watt-second, and up to and including 0.5 watt-second), and 1,000 volt capacitors - 120 percent of dc rated voltage.
Final measurements (at 25° C):
Insulation resistance:
Terminal to terminal - Not less than 30 percent of the value specified in table IV.
Capacitance: Within 10 percent of the initial value.
Dissipation factor:
Characteristic E - Change not more than 1 percent from initial value.
Characteristic F - Change not more than 1.5 percent from initial value.

Inspection lot: An inspection lot shall consist of capacitors of the same characteristic covered by this specification sheet.

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Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 11

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